

APPROVED



MSc Renewable Energy Engineering
School of Computing, Engineering and Technology

Version	3.0
Course Type	Postgraduate
Award Type	Master of Science - MSc
Awards	No Course Award Assigned
Code	0657
SCQF Level	Level 11
QAA Subject Benchmark	Engineering (2023)
SCQF Credit Points	180
ECTS Credit Points	90
Valid From	2026 September as of (September 2026)

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100175	energy engineering	energy engineering	100%
		Total	100%

Professional, Statutory and Regulatory Body (PSRB)
No PSRBs have been recorded

	The MSc Renewable Energy Engineering course aims to provide an in-depth understanding of practical and theoretical knowledge of modern renewable technologies in order to develop skilled renewable energy engineers that can support the renewable sector demands by 2030. The course is delivered in two modes full time through blended learning and part time through online learning via the University's Virtual Learning Environment (Moodle). Details of the modules contained in each semester can be seen further in this document. Regarding the course structure: Full Time Students For September entry, the sequence of modules will be: Semester 1 (taken in September), Semester 2 (taken in January), Semester 3 (project, taken in May)
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What the Course Involves	• For January entry, the sequence of modules will be: Semester 2 (taken in January), Semester 1 (taken in September), Semester 3 (project, taken in January) Part Time Students • For September entry, the sequence of modules will be: Semester 1 (taken in September), Semester 2 (taken in January), Semester 3 (taken in September), Semester 4 (taken in January), Semester 5 (project, taken in May and spanning 2 semesters) • For January entry, the sequence of modules will be: Semester 2 (taken in January), Semester 1 (taken in September), Semester 4 (taken in January), Semester 3 (taken in September), Semester 5 (project, taken in January and spanning 2 semesters) The course involves the completion of eight study modules followed by an individual MSc Project. Each module is assessed by coursework assignments and/or formal examination, and students are required to pass the 8 taught modules before progressing to the dissertation. Modules may be supplemented by industrial visits and industry speakers where practical and appropriate. On Campus learning will be at the forefront to allow extensive use of RGU's Moodle which facilitates interaction with teaching staff and other students for both full-time and part-time online learning delivery. Online delivery will mainly use the VLE Moodle and will be applied in concert with detailed lecture notes and activity questions. These activities will be carried out either as an on-line tutorial forum or through a set of training/learning events. Should sufficient demand arise, a module may be delivered as an in-class, face-to-face lecture and tutorial based course, with case study work, plus private study and discussion. Part-Time Online Learning The course is delivered through the RGU Moodle Virtual Learning Environment and there is no requirement to attend the University. The course content is identical to that of the full-time classroom-based course with the materials and sources of guided study being available online. The online library and, where applicable, eBook resources provide the same learning resources as for an on-campus student. The project will be delivered and supervised through online. For part-time online delivery the use of the Virtual Learning Environment, Moodle, will be the key and will be applied in concert with detailed lecture notes and activity questions. These activities will be carried out either as an on-line tutorial forum or through a set of training/learning events. Should sufficient demand arise, a module may be delivered as an in-class, face-to-face lecture and tutorial based course, with case study work, plus private study and discussion.
Course Collaborations	
No collaborations listed	
Introduction to the Course	This MSc Renewable Energy Engineering course is an interdisciplinary course across the School of Computing, Engineering & Technology and Aberdeen Business School. The course will provide graduates with practical experience as it incorporates practical activities, site visits, case study learning, and industry guest speakers lectures. There will also be an opportunity for completing the final project with industry, final projects will be supported by our School research groups working in these areas, thus providing graduates with research skills as well. The course also include group projects on real life applications that address various aspects of the UN Sustainable Development Goals. This course, based on the selected subjects in semester 2, allows students to graduate with their choice of route of specialisation to be either MSc in Renewable Energy Engineering with specialisation in "Innovation and Design" or MSc in Renewable Energy Engineering with specialisation in "Operation and Maintenance". This course aims to complement the renewable energy industry and research developments. This course is delivered blended learning full-time and online learning part-time.

Educational Aims of the Course	<u>This course aims to:</u> • Develop skilled renewable energy engineers that can meet the renewable energy sector expectations and increased needs. • Produce fit-for-purpose engineers with professional skills in the key areas of renewable energy systems design, maintenance, and operational management. • Provide students with real practical and technical skills which enhance their career opportunities, typically in all renewable energy industries, engineering firms and engineering service companies. • Provide students with independent investigation and research skills as well as presentation skills. Postgraduate Certificate Stage: Students learn the fundamentals and advances made in the technologies used within renewable energy systems. Postgraduate Diploma Stage: Students learn how to harvest, convert, store and deliver renewable energy as well as the computational methods used in renewable systems. With option 1 students gain extensive knowledge about the design and management of renewable energy systems; and with option 2 offers they gain extensive knowledge on the the operation and maintenance of renewable projects. Masters Stage: Students work on individual projects demonstrating their understanding and expertise of implementing the knowledge gained from Semester 1 and Semester 2 teachings into their project.
Awarding Institution	Robert Gordon University
Course Duration	Full-Time On-Campus Students Full-time 12 months with a September intake Part-time 16 months with a January intake Part-Time Online Students Part-time 28 months with a September intake Part-time 32 months with a January intake Students can choose to undertake a short-term (3 month) placement or a long-term (12 month) placement. All placements require approval from the School prior to commencement. If a student goes on placement, the duration of the course will be extended by the duration of the placement.
	The course will offer students a unique opportunity to gain in-depth knowledge and experience in renewable energy engineering in addition to skills required by industry such as project management and experimental work. Students will graduate with renewable energy project design skills or renewable energy project operation and maintenance skills, both required by many companies. By successfully completing the MSc Renewable Energy Engineering, graduates will have the advantage over other graduates from other renewable energy courses to apply for jobs due to their unique practical and research skills attained. Examples of jobs include: • Renewable energy project manager • Renewable design engineer

Distinctive Features of the Course	• Renewable research analyst • Renewable Project engineer • Renewable equipment manufacturers • Engineering and consulting companies specialist in any renewable energy projects The teaching of this course will involve gaining research expertise and research outputs will form part of the course in time. <u>Students will benefit from:</u> • On-site solar facility and hydrogen fuel cell visits. • Expert guest speaker lectures. • Practical activities. • Final Project industrial opportunities through RGU established links with various renewable energy companies.
Placement/Professional Experience	Students may elect to undertake a short placement or a year-long placement. The placement provides an opportunity to sample working life, be it in commerce, industry or the public sector, and apply knowledge and skills learned in the classroom to the challenges of the workplace. Students who have completed the placement as part of their course have a definite advantage when it comes to applying for graduate positions. With the combination of training, experience, and a successful degree, students are often offered subsequent employment by the placement provider with which they undertook their placement. While the student is on placement, they will be responsible to a designated member of the placement provider (the Industrial Supervisor) for the discharge of their day-to-day duties. Each student on placement is also assigned an Academic Tutor from the staff of the School, who will act as the student's personal tutor during the placement. The Academic Tutor will hold a small number of scheduled video calls with the student and their Industrial Supervisor during the placement. Students may choose to undertake a short placement. This normally comprises 3 months of continuous, full-time supervised work experience. The assessment for a short placement experience is based on a review of the student's work-related skills, conducted by the Industrial Supervisor. Students on a short placement will normally receive at least 1 video call with their Academic Tutor. As an alternative, students may undertake a year-long placement. The year-long industrial placement normally comprises of 12 months of continuous, full-time supervised work experience. The assessment for a year-long placement is based on a review of the student's work-related skills, conducted by both the student and their Industrial Supervisor. Students on a year-long placement will normally receive at least 2 video calls with their Academic Tutor. Placements must be undertaken after the completion of all taught modules and begin before the start of the final project. A student may choose to either undertake a placement at the same time as their final project or delay the start of their final project until the placement has finished. The course leader must approve this decision. The University provides support to students interested in undertaking a placement through the Placement Office and Centre for Employability and Community Engagement. The Placement Office maintains a pool of placement opportunities and establishes contact with new placement providers as necessary. It will normally assist students by informing them of suitable placement opportunities. Students are also given help and guidance by the Placement Office in obtaining their own placement from those advertised directly by companies or through their own connections. The Centre for Employability and Community Engagement offers further support for students, including with preparing their CV and developing interview skills.

	All placements require approval from the School prior to commencement, in order to ensure that they are suitable for the student and to confirm that the placement provider has put adequate supervisory and health-and-safety arrangements in place. Placements cannot be guaranteed and students wishing to undertake a placement are expected to reasonably engage with the process, for example by providing a copy of their CV to the Placement Office and by attending any interviews with potential placement providers when required. The credits awarded for the placement module are in addition to the 180 credits achievable from the other modules. Successful completion is not required for progression or graduation. The credits achieved for placement may not be used to compensate for any credits from the other modules.
Opportunities for Further Study	Following the completion of this MSc, further study of Doctor of Philosophy (PhD) or Engineering Doctorate (EngD) is possible.
External and Internal Indicators of Quality and Standards	Internal references include: • University Mission Statement External references include: • QAA Subject Benchmark for Engineering • SCQF Qualification Descriptors, Level 11 • Energy Institute Accreditation (planned for after first delivery of the course) • Peer review of staff research RGU's School of Computing, Engineering and Technology provides industry with highly qualified and committed graduates.
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
Course-Specific Entry Requirements	Applicants normally hold a 2:2 undergraduate honours degree, or its equivalent in Engineering or relevant Science discipline. Applicants whose first language is not English should have an IELTS of 6.5 overall, with no component lower than 5.5, or an equivalent.

Alternative Qualifications for Entry	Applicants with a lower degree classification and / or relevant work experience will be considered on a case by case basis.
Interview Procedures	Applicants are not usually interviewed, however, some may be interviewed if they do not meet the standard entry requirements.
Course-Specific Regulations	CEM100 cannot be condoned. All students, regardless of mode and start date, will typically undertake resits in the summer exam period. Exceptionally students who start in January may be offered a resit opportunity after completing two semesters, in April of the following year.
Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
University Strategy	The Purpose of the University is: - To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: - To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: - To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background.

	At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK. RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.
Course-Specific Learning Experience	The MSc Renewable Energy Engineering is designed to allow students to acquire specialist knowledge and generic skills in renewable energy within engineering, political and economic contexts. Graduates from the course have a greater advantage when applying for renewable energy jobs compared to graduates from the other generalised engineering courses. The MSc Renewable Energy Engineering will offer the student a unique experience and opportunity to gain in-depth knowledge in various renewable energy systems in addition to industrial skills required by the industry such as project management and experimental study. The course is suitable for graduates from a range of engineering and related disciplines and has been specifically designed to meet the high-growth global renewable energy industry. Students on this course are able to draw on a range of high-quality resources specific to renewable energy, such as a wide range of RGU Library resources including databases such as Web of Science, extensive journal subscriptions and a wide range of renewable energy related textbooks and e-books. The staff team is able to draw on their active experience with research and consultancy to provide teaching, which is relevant, up-to-date, and informed by real world experience. This course will enable the students to critically apply their thinking to effectively examine the theories and practices that make up the professional knowledge in Renewable Energy. Building on the students' first degree and the experience students have gained since graduating, this MSc course will allow the students to develop more advanced skills and knowledge in the field of renewable energy. The location of Robert Gordon University in Aberdeen, the energy capital of Europe, provides an accessible opportunity to involve specialists from the renewable energy and renewable industries. The School of Computing, Engineering and Technology and RGU have close links to a number of representatives within the local, national and international companies and local representatives from industry organisations such as the Energy Institute. The course is delivered in two modes, On Campus Learning Full-Time and Online Learning Part-Time. The part time online will utilise a range of innovative online teaching and learning methods to engage students. These will include online or recorded lectures, text-based open learning materials, online discussion forums and tutorials delivered by chat rooms and video technology. Delivery via Moodle provides considerable flexibility to students, enabling them to undertake the course from home or another setting, particularly suited to those working alongside the course. The participation of the student in the web-based activities is a key element of the learning process. As an online learner, the student is part of a virtual cohort, and communication and interaction among the cohort is a significant aspect of the learning process. Whilst Moodle is a collaborative learning environment, students are encouraged and supported to take responsibility for their own learning and wellbeing. This will allow the student to prepare for the project stage (if they progress to this), by becoming experienced in self-directed independent learning. Students are also strongly encouraged to attend locally arranged seminars and conferences such as those offered by the School. Lectures and tutorials

are combined with practical or computer-based simulation activities of industrial relevance. Students will also have guest speaker lecturers from industry representatives specialised in different fields of Renewable Energy. Students will also benefit from our exclusive wind tunnel, outdoor solar facility, hydrogen fuel cell and fully equipped IT labs.

The teaching and learning strategy for Renewable Energy Engineering Course aligns with that of the School of Computing, Engineering and Technology's teaching and learning strategy which aims to produce industrially relevant professional engineers. In line with the university's teaching, learning, and the student experience strategy, this will enable students to raise their aspirations and thrive in their professional careers as global citizens. RGU continues to lead the way in addressing global sustainability challenges and recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas. Consequently, sustainability is integrated into all courses at the School, equipping students with skills and competencies that are essential for their future careers. The teaching and learning strategy has been designed with reference to engineering council standards to ensure students develop the required professional competencies as they progress through the course. This includes a focus on sustainability with students tackling real-world engineering challenges whilst considering sustainable practice. The curriculum has been designed so that students develop their knowledge and understanding, practical skills, general cognitive skills, and key transferable skills at five levels corresponding to level 11 of the Scottish Credit and Qualifications framework.

Projects and teamwork are embedded in the course. This helps students to integrate theory and practical skills. It also encourages creativity and the development of transferable skills, including the ability to work in a team and an understanding of sustainable practice, enhancing students' employability. Project work, both in a team and individually, helps to develop students as independent, enthusiastic learners.

Whilst developing and delivering the curriculum the school of engineering has strong interaction with industry. This relevant and engaging curriculum enhances the teaching and learning experience, helping to ensure that content remains relevant to today's engineering challenges and improves students' employability.

The school of computing, engineering and technology recognises the importance of providing a high-quality learning environment with appropriate use of technology in teaching and learning. It encourages and supports staff in the development of innovative and effective teaching methods and excellent academic support. Student feedback is valued and used to develop and enhance the teaching and learning experience. Students develop the competencies required by a graduate engineer through a wide variety of teaching and assessment methods. These are aligned with the learning outcomes of the modules and the professional competencies students are expected to develop. These are described in more detail in the following sections.

Knowledge and Understanding

Knowledge and understanding are developed through a number of methods, which are module and delivery mode specific, such as:

- Lectures and tutorials - for the development of knowledge and understanding of a subject.
- Connect and Reflect Week Activity, Field Trip and/or Industrial Visit.
- Seminars, workshops, and/or guest lectures.
- Online discussion forums.

Those direct teaching methods are usually supported by coursework and project work, and are supplemented with direct study by means of:

- Learning materials on Virtual Learning Environment (Moodle).
- Textbooks and other recommended learning materials.
- Journal articles and conference papers.

Practice: Applied Knowledge and Understanding

Practical skills are developed through carrying out practical activities in an appropriate learning context. This is achieved by several methods, which are module and delivery mode specific, such as:

Teaching and Learning Strategy

- Practical exercises.
- Workshops.
- Individual and Group projects.
- Participation in field trips or industrial visits.

Generic Cognitive Skills

Generic cognitive skills such as critical analysis, synthesis, evaluation, and problem solving, are developed through active learning processes, which are module and delivery mode specific, such as:

- Group-learning activities - seminars, tutorials, or case studies.
- Computer-based simulation.
- Problem-solving exercises.
- Individual and group projects.
- Practical activities and engineering application work.

Generic cognitive Skills are also developed by means of project work.

Communication, ICT and Numeracy Skills

Communication, ICT and numeracy are developed through naturally arising opportunities within the curriculum, which are module and delivery mode specific. For instance:

- Written communication skills are developed through contributions to online discussion forums, developing answers to tutorial problems, coursework, and project reports.
- Oral communication skills are developed through interactions in lectures, tutorials and team activities, group projects, interactions and presentations in seminars and the presentations associated with projects.
- ICT skills are developed through the use of ICT applications and engineering software in learning and completing assignments.
- Numeracy Skills are developed through the interpretation, use and evaluation of a wide range of numerical and graphical data obtained through theoretical study, coursework, practical work, and projects.

Autonomy, Accountability and Working with others

Autonomy, accountability and working with others are skills that could be transferred to employment and other context freely. These skills are developed through naturally arising opportunities within the curriculum, which are module and delivery mode specific. For instance:

- Practical exercises.
- Engineering application activities.
- Group work.
- Case studies.
- Group projects.

Skills may also be developed through extra-curricular activities including:

- Work experience/placement.

	• Student representative work. • Engineering Society and club activities. • Social and cultural activities.
Assessment	The assessment of this course is guided by the course structure and a combination of different types of assessment is used, including project work, coursework, oral assessment, and closed book examinations. The type of the assessment used is determined by SCQF level, learning outcomes and nature of the module content. Knowledge and Understanding The primary mean of assessment of knowledge and understanding is by means of closed book examinations. However, it may also be assessed by means of: • Open book examinations. • Computer-based assessments. • Oral presentations. There is a greater emphasis on Knowledge and Understanding in the early stages of the course; this emphasis decreases as the course progresses when greater emphasis is placed on intellectual skills, particularly analysis, synthesis, evaluation, and problem solving. Practical Skills Practical skills are assessed by the most appropriate means, depending on the nature of the practical work. These skills are assessed by means of: • Practical exercises and reports for case studies or design exercises. • Computer based assessment (e.g. quiz and simulation). • Project work. Generic Cognitive Skills Cognitive skills include analysis, synthesis, evaluation, and problem solving. They are assessed by the following means: • Coursework (e.g. design exercises). • Case Studies • Project work (either of a group or individual nature). • Oral presentation. Communications, ICT and Numeracy Skills These are assessed by the following means: • Written reports (in the form of project reports and case studies). • Oral presentation (for projects, design exercises). students are expected to be able to communicate complex ideas in well-structured way (in both oral and written formats) using computer-based tools and are assessed on their ability to do this.

	Autonomy, Accountability and Working with others This category involves the skills which are readily transferable to employment and other contexts. They are assessed by the following means: • Coursework (for example, in group coursework assignments). • Group project submission.
University Support for Teaching, Learning and Assessment	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • a Team Around a Student (TAAS) system of pastoral support to students including online support www.rgu.ac.uk/policies [The TAAS School Student Support Model replaces the Personal Tutor model of student support from Session 2026-27]; • an extensive programme of student study skills delivered through Student Life and the Library; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in Scotland's Tertiary Enhancement Programme (STEP) www.step.ac.uk ; • support for staff from the Department of Academic Quality and Learning Enhancement (AQLE), including credit-rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • Moodle, http://campusmoodle.rgu.ac.uk , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
	The University provides a number of means of supporting teaching and learning: • RGU Assessment & Feedback Standard • RGU Blended Learning Standard • RGU Digital Learning Standard • student induction organised on a course or School basis; • an ongoing scheme of personal/pastoral support for students, including online support, provided by the <i>Inclusion Centre</i> • an extensive programme of student study skills delivered through <i>Student Life</i> and the <i>Library</i> • an extensive library of learning resources • close collaboration with industry and professional, statutory and regulatory bodies • the University's active participation in the Quality Assurance Agency's Quality Enhancement Themes, www.enhancementthemes.ac.uk ; • support for staff from the <i>Department for the Enhancement of Learning, Teaching and Assessment (DELTA)</i>, including credit rated teaching, learning and assessment provision • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities

Course-Specific Support for Teaching, Learning and Assessment

- access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub;
- Moodle, <http://campusmoodle.rgu.ac.uk/> , the University's dedicated virtual learning environment
- a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses
- the expanding provision of state-of-the-art, purpose-built facilities and buildings.

All students are offered the opportunity to take part in an Induction Programme which provides an introduction to the School, the staff and a chance to get to know the other students on the courses. This primarily takes place at the start of the course but there are also ongoing events in the first semester.

Further information is provided on the School's webpages and via the VLE, Moodle facility that enables a wide range of teaching and learning resources to be made available to students and members of staff. CampusMoodle also allows online marking and quick feedback to students.

All students are allocated a Personal Tutor. Personal Tutors support students with course related and personal issues and guide them in successfully completing the course.

Module Coordinators are available to support students with course and module related issues and concerns. Dedicated administrative support will be available to deal with any queries students may have.

Students will be supported through the School's integration of the institutional Team Around a Student (TAAS) approach, ensuring that every student has a point of contact who can provide advice, triage queries, and connect them with the most appropriate support and resources where issues may be affecting their studies or wider student experience.

The University has a Study Support Service. This provides support in the areas of academic writing, maths and statistics, IT support, English language and Study Skills. In addition to this, students have access to support on effective use of digital learning provided by the Digital Learning Team in Department of Academic Quality and Learning Enhancement (AQLE).

All students have access to a Student Handbook; this is provided in electronic format on the School's webpages. Students undertaking individual projects are provided with a detailed Project Handbook covering all aspects of the project operation and assessment. Extensive written guidance is also available via the VLE. All students have an account on the University computer network and can access PCs in the school and across the University. The support staff also assist students with administrative queries.

University Monitoring of Quality and Standards

The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including:

- *Annual Course Appraisal Reports* are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires;
- *Institution-Led Subject Review* , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process;
- *External Examiner Annual Reports* ;
- Ongoing liaison with industrial/professional liaison groups.

Formal Committees with responsibility for monitoring and evaluating quality and standards:

- Staff/Student Liaison Committees, or equivalent;

	• Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three-year interim review to monitor progress against actions/issues raised through the review process • <i>External Examiner Annual Reports</i> • Ongoing liaison with industrial/professional liaison groups Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent • Course/Programme Management Teams • School Academic Boards • Assessment Boards • Quality Assurance and Enhancement Committee and associated Sub-Committees Annual Course Appraisals are reviewed and approved by the Course Management Team and School Academic Board which meet at least once per semester. The Course Management Team considers teaching and learning issues arising from feedback generated by the External Examiner Annual Reports, Staff and Student Liaison Committees, Student Experience Questionnaires and the National Student Survey. The Course Management Team and School Academic Board also review all proposed changes to the structure and content of courses. The School's Teaching, Learning & Assessment Committee regularly looks at metrics such as student experience and student achievement to ensure that the courses are continuing to deliver the required standard set by the University. It also supports the ongoing enhancement of approaches to teaching, learning and assessment. The quality and standard of the assessment are ensured through a robust moderation process in the School. All summative assessments are moderated by an academic staff and an external examiner. The marking of all assessments is completed in full compliance with the University's Assessment policy.
	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements.

University Student Feedback Mechanisms	This approach is underpinned by the <i>Student Partnership Agreement</i> which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: Student Representation & Partnership RGU . There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience. Mechanisms through which students are engaged and supported include: • participation in student evaluation questionnaires and where relevant the <i>National Student Survey (NSS)</i> ; • staff/student liaison arrangements at course/programme level; • support for Student Representatives; • student representation on Institution-Led Subject Review Panels and Validations; • student representation on key University committees.
Course-Specific Student Feedback Mechanisms	The University and RGU Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements. This approach is underpinned by the <i>Student Partnership Agreement</i> which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership CampusMoodle Area: www.rgu.ac.uk/studentpartnership . There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience. Mechanisms through which students are engaged and supported include: • participation in student evaluation questionnaires and where relevant the <i>National Student Survey (NSS)</i> • staff/student liaison arrangements at course/programme level • support for Student Representatives • student representation on Institution-Led Subject Review Panels and Validations • student representation on key University committees The Staff Student Liaison Committee meets at least once per semester. The membership of the Committee consists of the Course Leaders, course administrators and student representatives. Student representatives are asked to collect comments from their class and they are also encouraged to liaise with Course Leaders/Year Tutors when issues arise (rather than waiting for the next Staff Student Liaison Committee meeting). Minutes of the meetings are posted on Moodle for all students on the course to access.

Awards
Stage 1

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Postgraduate Certificate - PgCert	Renewable Energy Engineering	Aberdeen, United Kingdom	Exit Award		60
Postgraduate Diploma - PgDip	Renewable Energy Engineering	Aberdeen, United Kingdom	Exit Award		120
Master of Science - MSc	Renewable Energy Engineering (Operations and Maintenance)	Aberdeen, United Kingdom	Target Award		180
Master of Science - MSc	Renewable Energy Engineering (Design and Innovation)	Aberdeen, United Kingdom	Target Award		180

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	Design solutions for complex problems that evidence some originality and meet a combination of societal, customer and renewable energy industries engineering demands.	SC1 SC2	SCQF 11, SCQF 11
LLO2	Apply a comprehensive knowledge of mathematics, natural science and engineering principles to the solution of complex renewable energy systems problems.	SC1 SC2 SC4	SCQF 11, SCQF 11, SCQF 11
LLO3	Select and apply appropriate computational and analytical techniques to model renewable energy projects.	SC1 SC3 SC4	SCQF 11, SCQF 11, SCQF 11
LLO4	Formulate and analyse complex renewable energy projects problems within the context of broader disciplines of science, engineering, management and business to reach substantiated conclusions.	SC1 SC5	SCQF 11, SCQF 11
LLO5	Select and critically evaluate technical literature, other sources of information, and research process to solve complex renewable energy systems problems.	SC1 SC2 SC3 SC5 SC4	SCQF 11, SCQF 11, SCQF 11, SCQF 11, SCQF 11
LLO6	Communicate effectively complex renewable engineering projects with technical and non-technical audiences evaluating the effectiveness of the used methods through professional project reports as well as confident, high-quality oral presentations in a range of contexts areas covered in the course.	SC1 SC2 SC5 SC4	SCQF 11
LLO7	Evaluate the environmental and societal impact of renewable energy systems through the assessment of their products or process entire life-cycle in order to minimize adverse impacts.	SC2 SC5	SCQF 11, SCQF 11
LLO8	Use practical laboratory and workshop skills to investigate complex renewable energy systems problems	SC2	SCQF 11
LLO9	Apply an integrated or systems approach to the solution of complex renewable energy systems problems using the full range of problem solving, communication, information retrieval, and the effective use of general Information Technology skills covered in course	SC2 SC4	SCQF 11, SCQF 11
LLO10	Function effectively as an individual or as a member or leader of a team (with minimal guidance where appropriate), confident and accurate selection and application of principles and procedures appropriate to the resolution of a range of renewable projects situations and professional problems within the discipline areas covered in the course.	SC3 SC5 SC4	SCQF 11, SCQF 11, SCQF 11
LLO11	Evaluate the effectiveness of own and team performance through evaluating theories, processes solutions and outcomes critically and effectively;	SC3 SC5	SCQF 11, SCQF 11
LLO12	Design innovative solutions for complex renewable energy systems problems that evidence creatively, drive and change.	SC3	SCQF 11

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
		SCQF 7 - SCQF 7 SCQF 8 - SCQF 8

SC2	Practice: Applied knowledge, skills and understanding	SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC5	Autonomy, accountability and working with others	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

Delivery Range

<i>Title</i>	<i>Description</i>
Full Time On Campus	Route A - Design and Innovation: Non-Placement Route B - Design and Innovation: Short Placement Route C - Design and Innovation: Long Placement Route D - Operation and Maintenance: Non-Placement Route E - Operation and Maintenance: Short Placement Route F - Operation and Maintenance: Long Placement
Part Time Online	Route A - Design and Innovation: Non-Placement Route B - Design and Innovation: Short Placement Route C - Design and Innovation: Long Placement Route D - Operation and Maintenance: Non-Placement Route E - Operation and Maintenance: Short Placement Route F - Operation and Maintenance: Long Placement

Stage 1 / Semester 1**Full Time On Campus**

Route A - Design and Innovation: Non-Placement Route B - Design and Innovation: Short Placement Route C - Design and Innovation: Long Placement Route D - Operation and Maintenance: Non-Placement Route E - Operation and Maintenance: Short Placement Route F - Operation and Maintenance: Long Placement

Part Time Online

Route A - Design and Innovation: Non-Placement Route B - Design and Innovation: Short Placement Route C - Design and Innovation: Long Placement Route D - Operation and Maintenance: Non-Placement Route E - Operation and Maintenance: Short Placement Route F - Operation and Maintenance: Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CBM204	Data Analytics for Business Decisions		Sandra Cannon (sc11706)	Yes	3.0	15	Level 11
Full Time On Campus	BSM158	Renewable Energy Management		Ayodele Asekomeh (aa7163)	Yes	1.0	15	Level 11
Full Time On Campus	ENM275	Renewable Energy Technologies		Dallia Ali (da4630)	Yes	2.0	15	Level 11
Part Time	ENM275	Renewable Energy Technologies		Dallia Ali	Yes	2.0	15	Level

Online				(da4630)				11
Full Time On Campus	ENM509	Wind Energy Systems		Ibiye Iyalla (ii5282)	Yes	2.0	15	Level 11
Part Time Online	ENM509	Wind Energy Systems		Ibiye Iyalla (ii5282)	Yes	2.0	15	Level 11

Stage 1 / Semester 2

Full Time On Campus

Route A - Design and Innovation: Non-Placement Route B - Design and Innovation: Short Placement Route C - Design and Innovation: Long Placement Route D - Operation and Maintenance: Non-Placement Route E - Operation and Maintenance: Short Placement Route F - Operation and Maintenance: Long Placement

Part Time Online

Route A - Design and Innovation: Non-Placement Route B - Design and Innovation: Short Placement Route C - Design and Innovation: Long Placement Route D - Operation and Maintenance: Non-Placement Route E - Operation and Maintenance: Short Placement Route F - Operation and Maintenance: Long Placement

Core								
Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
Full Time On Campus	ENM271	Advanced Power Systems For Renewables' Integration	Routes A, B, C	Dallia Ali (da4630)	Yes	2.0	15	Level 11
Full Time On Campus	CEM101	Innovation and Sustainability for Global Impact	Routes A, B, C, D, E, F	Ruissein Mahon (rm8781)	Yes	2.0	15	Level 11
Part Time Online	CEM101	Innovation and Sustainability for Global Impact		Ruissein Mahon (rm8781)	Yes	2.0	15	Level 11
Full Time On Campus	ENM273	Inspection and Maintenance	Routes D, E, F	Anil Prathuru (ap7527)	Yes	2.0	15	Level 11
Part Time Online	CEM201	Introduction to Solar Energy Systems		Mark Zarb (mz7371)	Yes	2.0	15	Level 11
Full Time				Mark Zarb				Level

On Campus	CEM201	Introduction to Solar Energy Systems	Routes A, B, C, D, E, F	(mz7371)	Yes	2.0	15	11
Full Time On Campus	ENM274	Reliability and Life Cycle Analysis	Routes D, E, F	Murphy Lai (ml12019)	Yes	3.0	15	Level 11
Full Time On Campus	ENM272	Renewable Energy System Design	Routes A, B, C	Ityona Amber (ia11260)	Yes	2.0	15	Level 11

Stage 1 / Semester 3**Full Time On Campus**

Route A - Design and Innovation: Non-Placement Route B - Design and Innovation: Short Placement Route C - Design and Innovation: Long Placement Route D - Operation and Maintenance: Non-Placement Route E - Operation and Maintenance: Short Placement Route F - Operation and Maintenance: Long Placement

Part Time Online

Route A - Design and Innovation: Non-Placement Route B - Design and Innovation: Short Placement Route C - Design and Innovation: Long Placement Route D - Operation and Maintenance: Non-Placement Route E - Operation and Maintenance: Short Placement Route F - Operation and Maintenance: Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Part Time Online	CBM204	Data Analytics for Business Decisions		Sandra Cannon (sc11706)	Yes	3.0	15	Level 11
Full Time On Campus	CEM100	MSc Project	Routes A, B, C, D, E, F	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
Part Time Online	BSM158	Renewable Energy Management		Ayodele Asekomeh (aa7163)	Yes	1.0	15	Level 11
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CEM104	Short Postgraduate Placement	Routes B, E	Mark Bartlett (mb9400)	No	1.0	15	Level 11
Full Time On Campus	CEM105	Year-long Postgraduate Placement	Routes C, F	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Stage 1 / Semester 4**Part Time Online**

Route A - Design and Innovation: Non-Placement Route B - Design and Innovation: Short Placement Route C - Design and Innovation: Long Placement Route D - Operation and Maintenance: Non-Placement Route E - Operation and Maintenance: Short Placement Route F - Operation and Maintenance: Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>

Part Time Online	ENM271	Advanced Power Systems For Renewables' Integration	Routes A, B, C	Dallia Ali (da4630)	Yes	2.0	15	Level 11
Part Time Online	ENM273	Inspection and Maintenance	Routes D, E, F	Anil Prathuru (ap7527)	Yes	2.0	15	Level 11
Part Time Online	ENM274	Reliability and Life Cycle Analysis	Routes D, E, F	Murphy Lai (ml12019)	Yes	3.0	15	Level 11
Part Time Online	ENM272	Renewable Energy System Design	Routes A, B, C	Ityona Amber (ia11260)	Yes	2.0	15	Level 11

Stage 1 / Semester 5**Part Time Online**

Route A - Design and Innovation: Non-Placement Route B - Design and Innovation: Short Placement Route C - Design and Innovation: Long Placement Route D - Operation and Maintenance: Non-Placement Route E - Operation and Maintenance: Short Placement Route F - Operation and Maintenance: Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Part Time Online	CEM100	MSc Project	Routes A, B, C, D, E, F. Undertaken over two sem.	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Part Time Online	CEM104	Short Postgraduate Placement	Routes B, E	Mark Bartlett (mb9400)	No	1.0	15	Level 11
Part Time Online	CEM105	Year-long Postgraduate Placement	Routes C, F	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Course Deliveries

<i>Stage</i>	<i>Semester</i>	<i>Delivery Type</i>	<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
1	Semester 1	Core	Full Time On Campus	CBM204	Data Analytics for Business Decisions		Sandra Cannon (sc11706)	Yes	3.0	15	Level 11
1	Semester 1	Core	Full Time On Campus	BSM158	Renewable Energy Management		Ayodele Asekomeh (aa7163)	Yes	1.0	15	Level 11
1	Semester 1	Core	Full Time On Campus	ENM275	Renewable Energy Technologies		Dallia Ali (da4630)	Yes	2.0	15	Level 11
1	Semester 1	Core	Part Time Online	ENM275	Renewable Energy Technologies		Dallia Ali (da4630)	Yes	2.0	15	Level 11
			Full								

1	Semester 1	Core	Time On Campus	ENM509	Wind Energy Systems		Ibiye Iyalla (ii5282)	Yes	2.0	15	Level 11
1	Semester 1	Core	Part Time Online	ENM509	Wind Energy Systems		Ibiye Iyalla (ii5282)	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	ENM271	Advanced Power Systems For Renewables' Integration	Routes A, B, C	Dallia Ali (da4630)	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	CEM101	Innovation and Sustainability for Global Impact	Routes A, B, C, D, E, F	Ruissein Mahon (rm8781)	Yes	2.0	15	Level 11
1	Semester 2	Core	Part Time Online	CEM101	Innovation and Sustainability for Global Impact		Ruissein Mahon (rm8781)	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	ENM273	Inspection and Maintenance	Routes D, E, F	Anil Prathuru (ap7527)	Yes	2.0	15	Level 11
1	Semester 2	Core	Part Time Online	CEM201	Introduction to Solar Energy Systems		Mark Zarb (mz7371)	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	CEM201	Introduction to Solar Energy Systems	Routes A, B, C, D, E, F	Mark Zarb (mz7371)	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	ENM274	Reliability and Life Cycle Analysis	Routes D, E, F	Murphy Lai (ml12019)	Yes	3.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	ENM272	Renewable Energy System Design	Routes A, B, C	Ityona Amber (ia11260)	Yes	2.0	15	Level 11
1	Semester	Core	Part Time	CBM204	Data Analytics for Business Decisions		Sandra Cannon	Yes	3.0	15	Level

	3		Online				(sc11706)				11
1	Semester 3	Core	Full Time On Campus	CEM100	MSc Project	Routes A, B, C, D, E, F	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
1	Semester 3	Core	Part Time Online	BSM158	Renewable Energy Management		Ayodele Asekomeh (aa7163)	Yes	1.0	15	Level 11
1	Semester 3	Elective	Full Time On Campus	CEM104	Short Postgraduate Placement	Routes B, E	Mark Bartlett (mb9400)	No	1.0	15	Level 11
1	Semester 3	Elective	Full Time On Campus	CEM105	Year-long Postgraduate Placement	Routes C, F	Mark Bartlett (mb9400)	No	1.0	120	Level 11
1	Semester 4	Core	Part Time Online	ENM271	Advanced Power Systems For Renewables' Integration	Routes A, B, C	Dallia Ali (da4630)	Yes	2.0	15	Level 11
1	Semester 4	Core	Part Time Online	ENM273	Inspection and Maintenance	Routes D, E, F	Anil Prathuru (ap7527)	Yes	2.0	15	Level 11
1	Semester 4	Core	Part Time Online	ENM274	Reliability and Life Cycle Analysis	Routes D, E, F	Murphy Lai (ml12019)	Yes	3.0	15	Level 11
1	Semester 4	Core	Part Time Online	ENM272	Renewable Energy System Design	Routes A, B, C	Ityona Amber (ia11260)	Yes	2.0	15	Level 11
1	Semester 5	Core	Part Time Online	CEM100	MSc Project	Routes A, B, C, D, E, F. Undertaken over two sem.	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
1	Semester 5	Elective	Part Time Online	CEM104	Short Postgraduate Placement	Routes B, E	Mark Bartlett (mb9400)	No	1.0	15	Level 11
1	Semester 5	Elective	Part Time Online	CEM105	Year-long Postgraduate Placement	Routes C, F	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
Aberdeen, United Kingdom	On-Campus	Full Time	
All Applicable Locations	Online	Part Time	

Sources of Further Information

No sources of further information recorded
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